

2. Common Elements Between Two Arrays

Aim

To write a Python program to find the **common elements between two arrays** and analyze its time complexity.

Procedure

1. Create two arrays containing integer elements.
2. Create an empty list to store the common elements.
3. Traverse the first array using a **for** loop.
4. Check whether each element is present in the second array.
5. If the element is present, add it to the common elements list.
6. Avoid adding duplicate elements to the result.
7. Display the common elements.
8. Analyze the time complexity of the algorithm.



```
main.py Visualize Run ▶ Output
1 def commonElements(arr1, arr2):
2     common = []
3     for element in arr1:
4         if element in arr2:
5             if element not in common:
6                 common.append(element)
7     return common
8 arr1 = [1, 2, 3, 4, 5]
9 arr2 = [3, 4, 5, 6, 7]
10 result = commonElements(arr1, arr2)
11 print("Common Elements:", result)
```

Common Elements: [3, 4, 5]

Result

Thus, the Python program successfully finds the **common elements between the two given arrays**.